

in specially-architected greenhouses. It can even be done in a common room, on the terrace or basement of an apartment complex, office or factory, just growing enough to feed the folks who live or work there.

However, if you look at the time-pressed urban population, they will hardly be motivated to implement vertical farming if its maintenance is too tiresome. So Sanjay Aggarwal, founder and chief executive officer of Clover Organic, a pioneer in vertical farming in India, suggests, "It would make sense for a person to sell the infrastructure and then charge a monthly fee for looking after the infrastructure."

This kind of facility would also be very helpful for a traditional farmer who decides to try out vertical farming. The farmer might be great at farming, but might not know technology. So if you are able to install and maintain the tech infrastructure, the farmer can put his farming knowledge to great use.

Set an example and sell the tech. Factory farming seems to be the fad now. Tech industry majors like Toshiba, Sharp and Fujitsu are all into vertical farming. Fujitsu grows lettuce in a former silicon chip manufacturing plant, and Sharp grows strawberries in Dubai.

Interestingly, Fujitsu's facility is less than 100km away from the location of the 2011 Fukushima nuclear disaster, and it grows 3000 heads of lettuce a day, which sell for three times the price of other lettuce.

Toshiba has a high-tech vertical farm near Tokyo where it grows lettuces—it projects that the produce will be around three million heads of lettuce per year. Basically, these vertical farming setups have a two-fold benefit. Not only can the



Fig. 2: Each crop has an ideal light recipe for best quality, taste and yield (Image courtesy: Philips)

produce be sold in the market, but once the systems are fine-tuned, these can be sold to several vertical farms, worldwide.

Tech based enablers. There are technologies that can help all kinds of vertical farmers, whether or not they use technology to control the environment. If you can think up innovative technologies that can help this segment, say a mobile-enabled sensor network to monitor the nutrient levels or a mobile app to exclusively shop for vertical farming needs, then you have an existing market to tap, and might get quicker returns.

Raising money to raise crops

In India, tech based vertical farming is a greenfield. Internationally, startups in this space are hopeful of being very profitable, but most are still in their first year of operation. So there is no clear business model for you to follow; no black-and-white instruction on what is right or what is wrong.

Here are some tips gleaned from here and there, for those who wish to foray into this unexplored terrain:

1. If you are developing off-the-shelf products for vertical farming, you can start small by crowd-sourcing funds. After developing one or two successful products, you can expand your product range later with the profits earned from initial

products. Greenopia, for instance, raised funds on Wishberry. They crossed their funding goal in just two months, raising ₹ 958,500, contributed by 17 backers from six countries. Their campaign was backed by popular names like Rajan Anandhan, managing director and vice president of Google India, Sarad Sharma, former chief executive officer of Yahoo India,

Dori Tunstall, a renowned professor in design anthropology, Diana Greenberg, a well-known user experience designer, and Margaret Atwood, the famous writer.

2. If you want to do vertical farming yourself, in a tech-intensive setup, a small terrace area will not make sense because the volume of business will not justify the heavy investment. You need to go for a larger infrastructure, preferably situated close to a city.

Learning from their past mistakes, Paul Lightfoot, chief executive of BrightFarms, stated in a *Wall Street Journal* story that, "Building farms on city rooftops is a foolish endeavour because of higher costs and additional time for permitting." The firm, which has raised more than US\$ 25 million in equity and more than US\$ 15 million in project finance, is now focusing on greenhouse farms in locations outside of urban centres. Speaking of city terrace based farms, Lightfoot says, "You would scale very slowly, and waste investors' money."

3. AeroFarms, one of the global pioneers in this space, has raised US\$ 70 million in corporate and project financing. The company is not yet profitable, but expects to become cash-flow positive in its first year of operation. In a financial news report last year, chief executive officer